NoSQL

Query processing engine: Prestodb

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SoICT - HUST

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History

- 2012 Fall: Project started at Facebook
  - Designed for interactive query
  - with speed of commercial data warehouse
  - and scalability to the size of Facebook
- 2013 Winter: Open sourced
  - 30+ contributes in 6 months
  - including people from outside of Facebook
- 2019: 300+ contributors

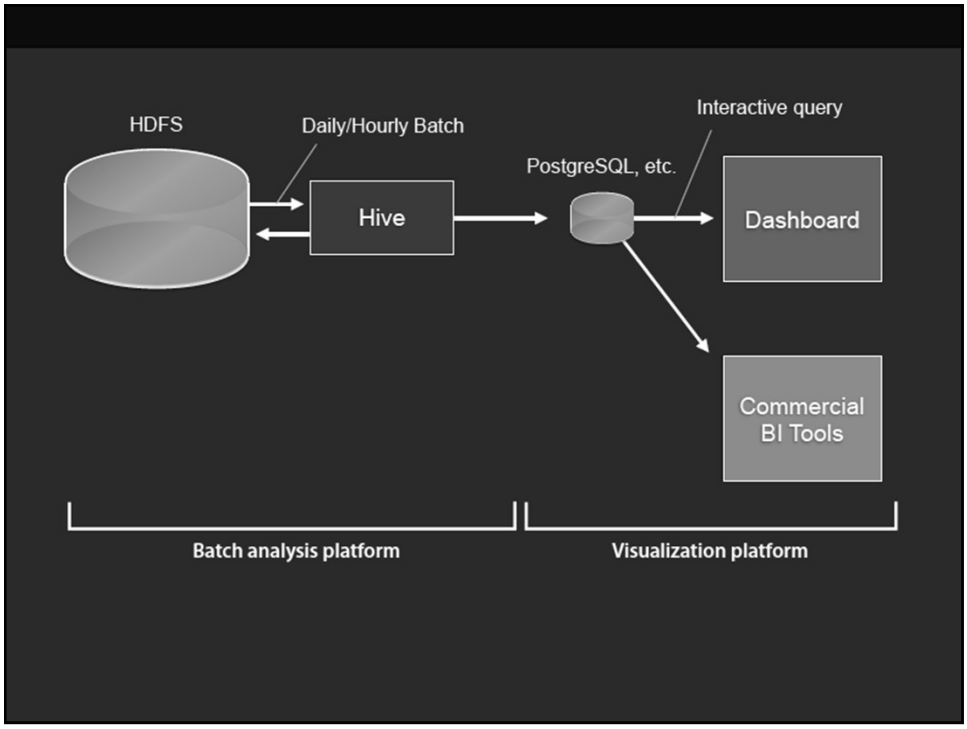
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Motivation

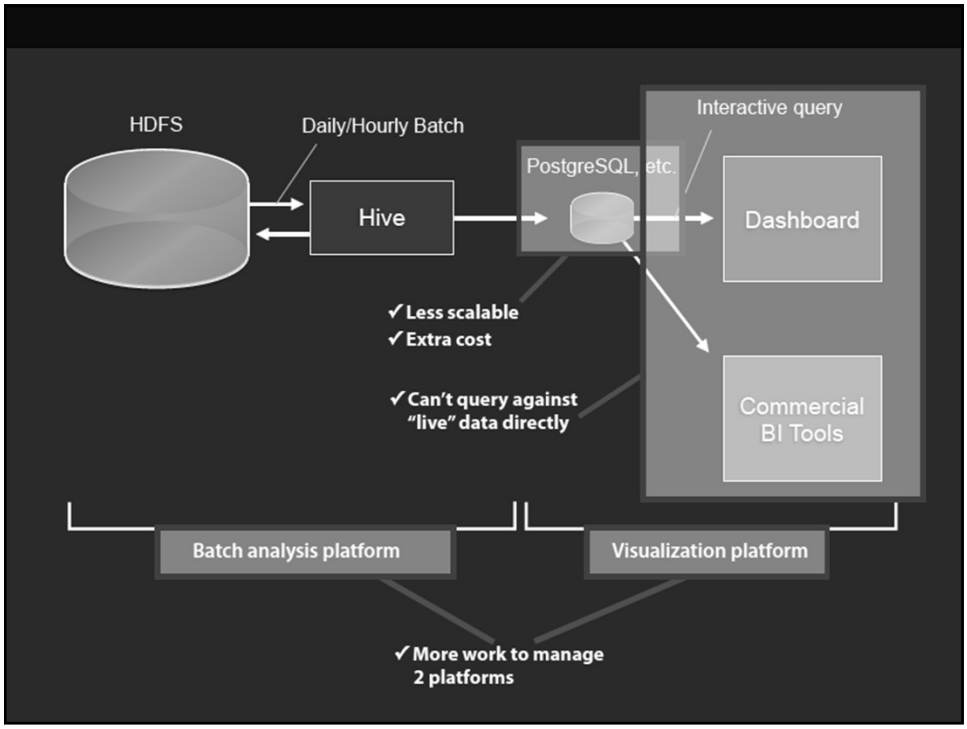
- We couldn't visualize data in HDFS directly using dashboards or BI tools
  - because Hive is too slow (not interactive)
  - or ODBC connectivity is unavailable/unstable
- We needed to store daily-batch results to an interactive DB for quick response (PostgreSQL, Redshift, etc.)
  - Interactive DB costs more and less scalable by far
- Some data are not stored in HDFS
  - We need to copy the data into HDFS to analyze

ability to quickly and easily extract insights from large amounts of data

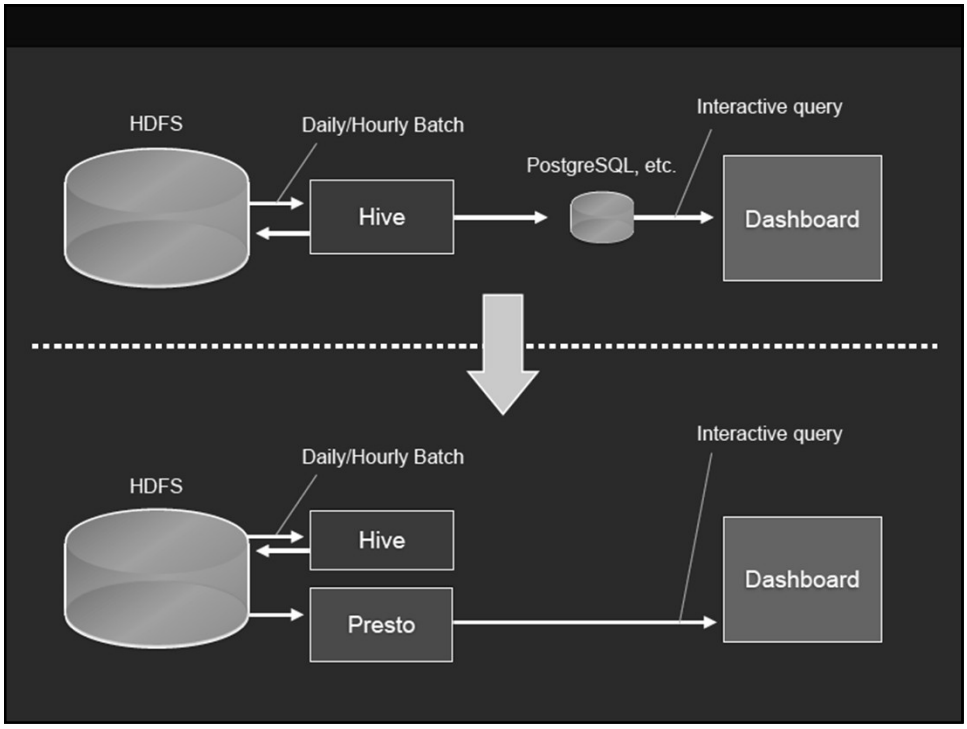
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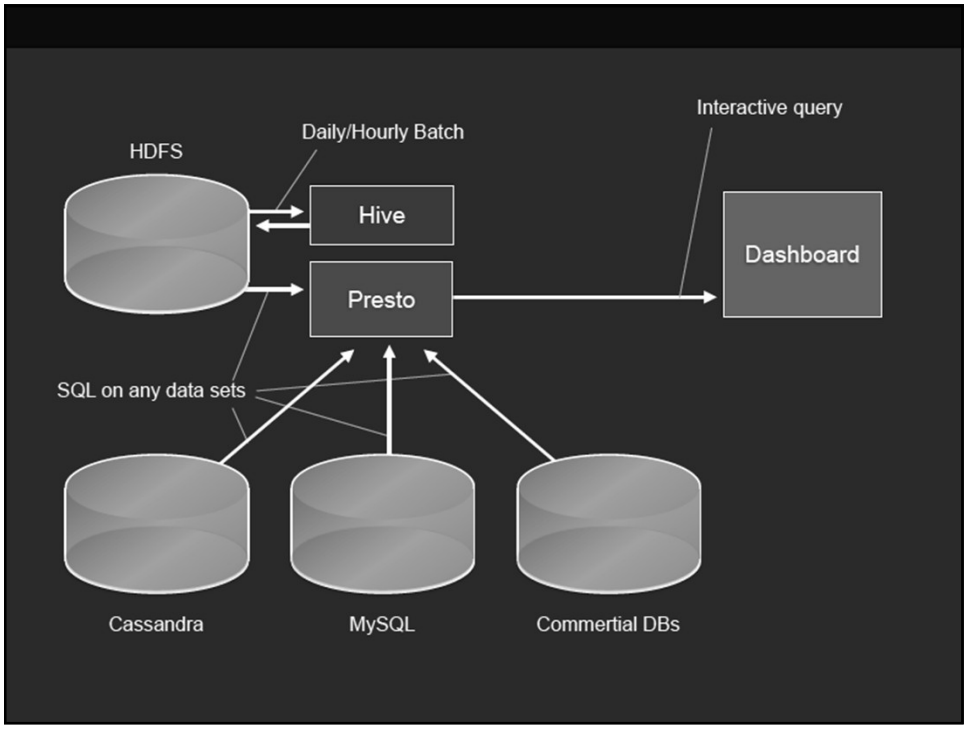
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HDFS
Daily/Hourly Batch
Hive
PostgreSQL, etc.
Interactive query
Dashboard
✓Less scalable
✓Extra cost
✓Can't query against "live" data directly
Commercial BI Tools
Batch analysis platform
Visualization platform
✓More work to manage 2 platforms

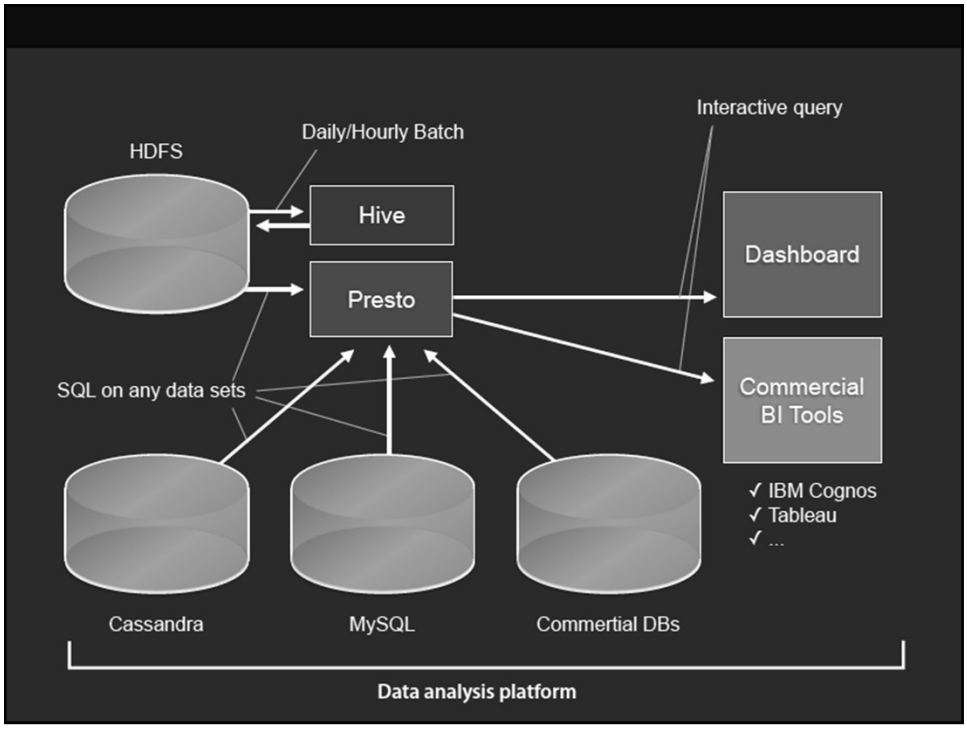
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HDFS
Daily/Hourly Batch
Hive
PostgreSQL, etc.
Interactive query
Dashboard
HDFS
Daily/Hourly Batch
Hive
Presto
Interactive query
Dashboard

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HDFS
Daily/Hourly Batch
Interactive query
Hive
Presto
Dashboard
SQL on any data sets
Cassandra
MySQL
Commertial DBs

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HDFS
Daily/Hourly Batch
Interactive query
Hive
Presto
Dashboard
Commercial BI Tools
SQL on any data sets
✓ IBM Cognos
✓ Tableau
✓ ...
Cassandra
MySQL
Commertial DBs
Data analysis platform

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What Presto can do?

- Open-source distributed SQL query engine that has run in production at Facebook since 2013
  - ANSI SQL interface
- Query interactively (in milli-seconds to minutes)
  - MapReduce and Hive are still necessary for ETL
- Query using commercial BI tools or dashboards
  - Reliable ODBC/JDBC connectivity
- Query across multiple data sources such as Hive, HBase, Cassandra, or even commercial DBs
  - Plugin mechanism
- Integrate batch analysis + visualization into a single data analysis platform

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Presto deployment

- Facebook (2013)
  - Multiple geographical regions
  - Scaled to 1,000 nodes
  - Actively used by 1,000+ employees who run 30,000+ queries every day
  - Processing 1PB/day







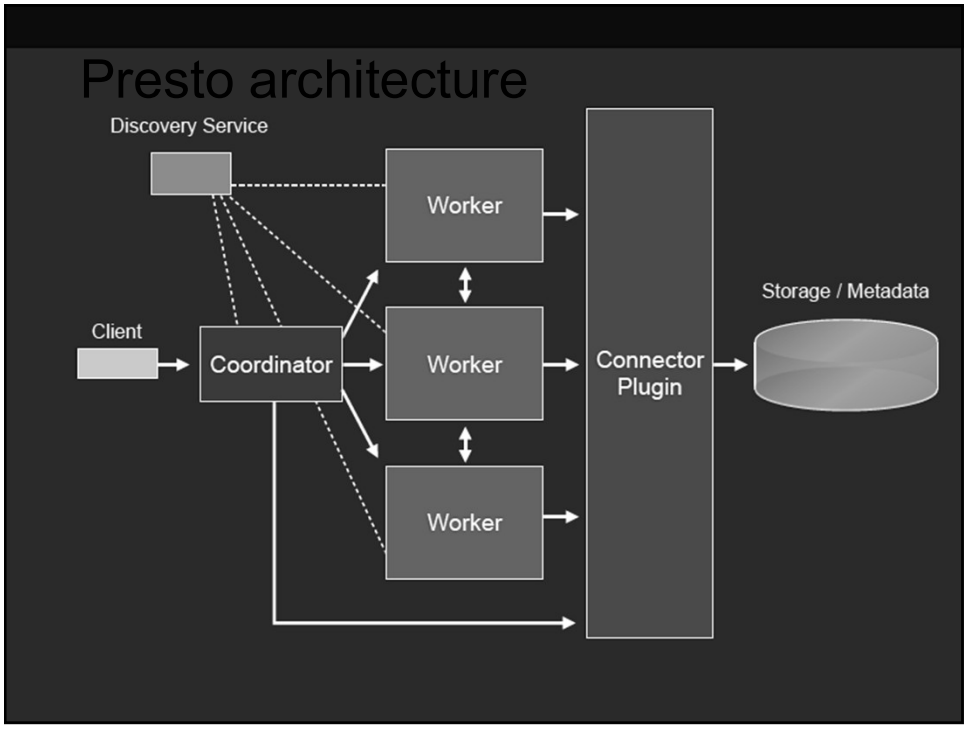




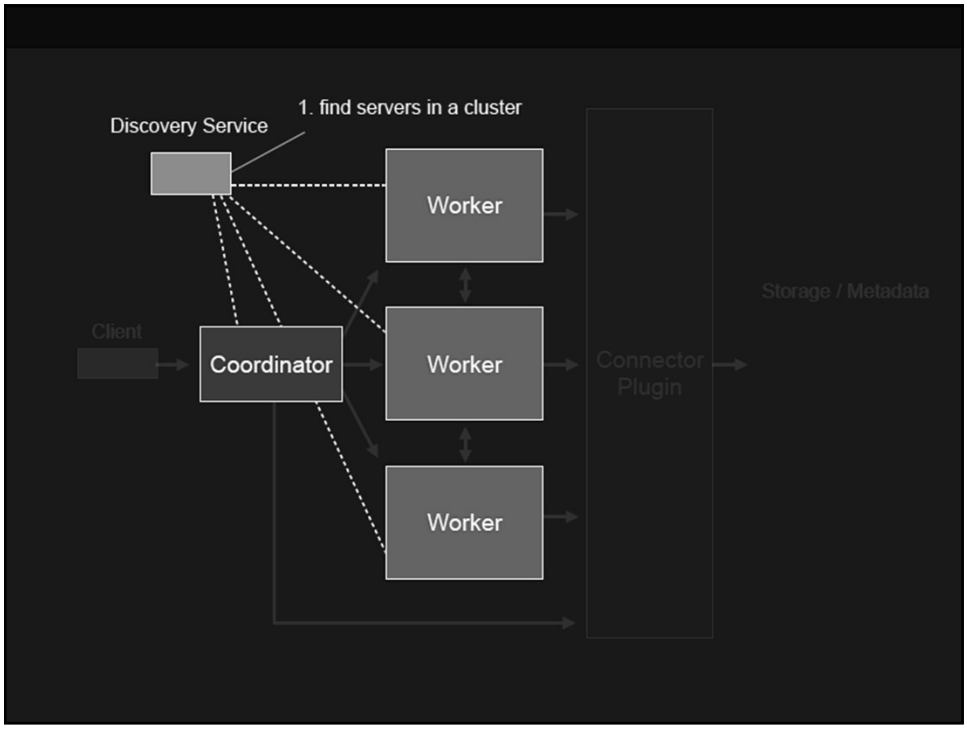
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zuora
YAHOO! JAPAN
JD.COM 京东
FreeWheel
amazon
Marin SOFTWARE
mercado Libre
LinkedIn
jampp
looker
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WIX
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airbnb
Walmart
Alibaba Group
UBER
slack
Bloomberg
GROUPON
GREE
cogo labs
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shopify
Atlassian
Pinterest
SHAZAM
openspan
NASDAQ
Dropbox
CUEBIQ
NETFLIX

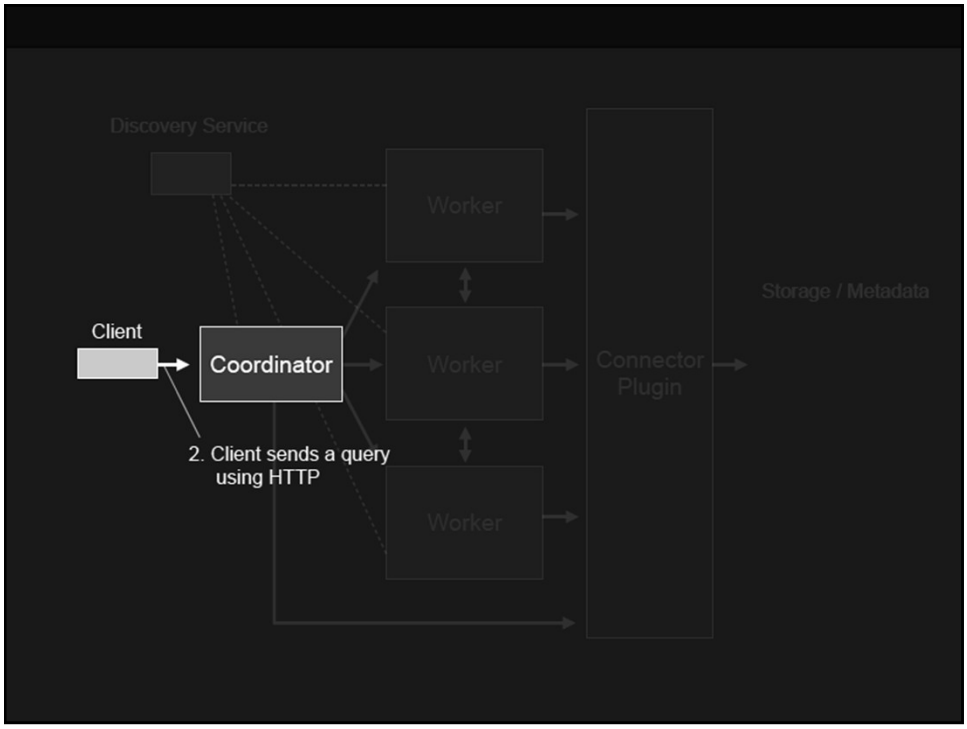
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Presto architecture
Discovery Service
Worker
Worker
Worker
Client
Coordinator
Connector Plugin
Storage / Metadata

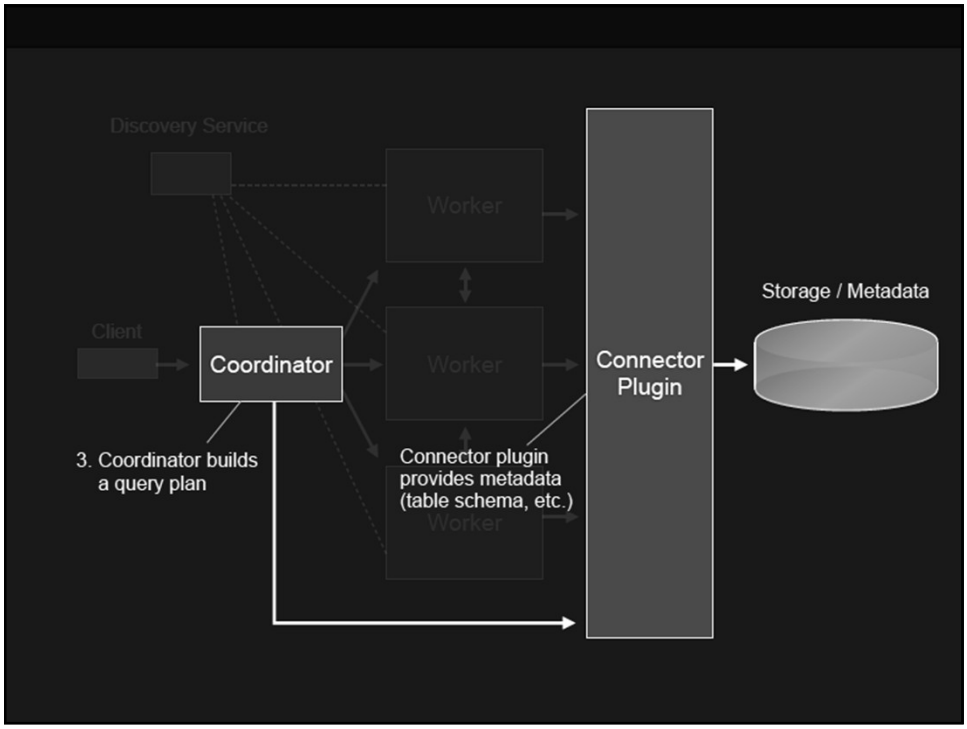
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Discovery Service
1. find servers in a cluster
Worker
Storage / Metadata
Client
Coordinator
Worker
Connector Plugin
Worker

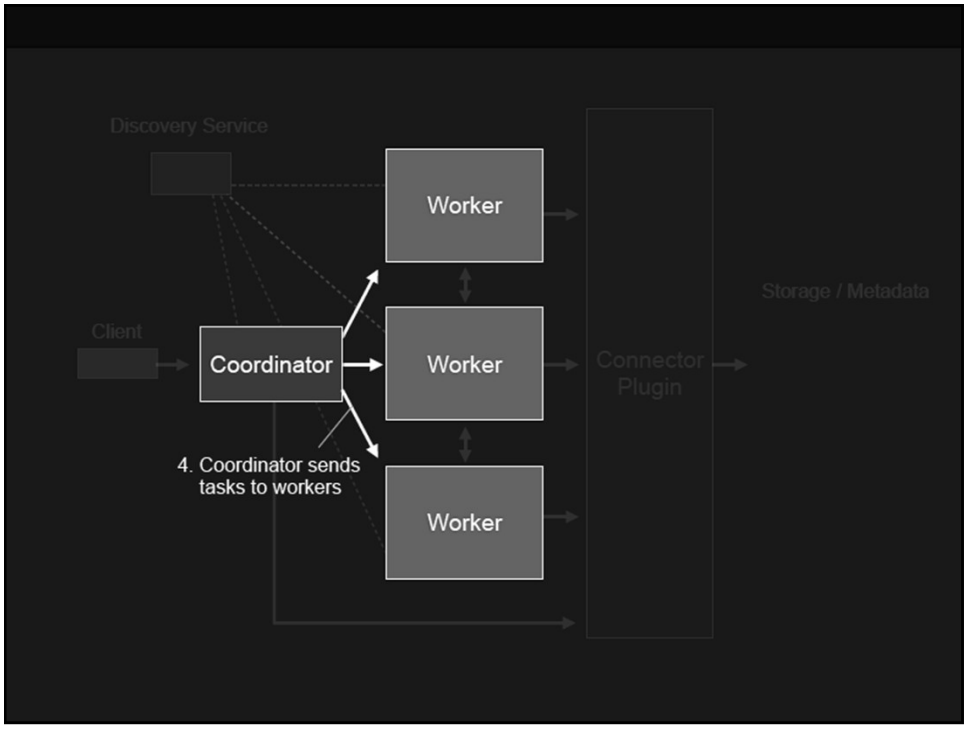
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Discovery Service
Worker
Storage / Metadata
Client
Coordinator
Worker
Connector Plugin
2. Client sends a query using HTTP
Worker

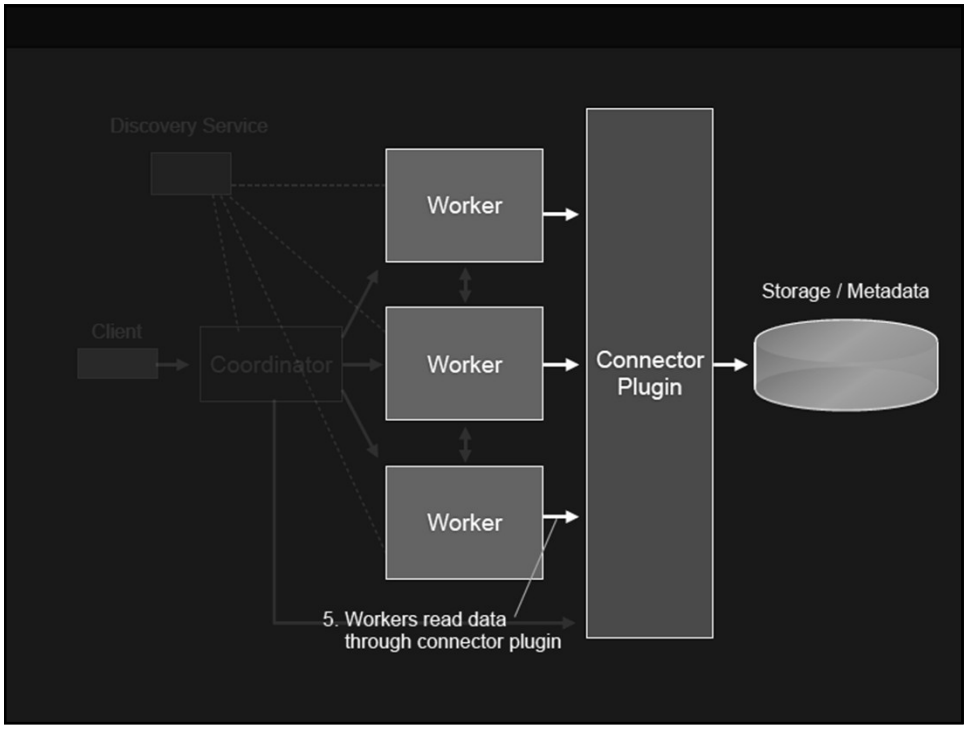
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Discovery Service
Worker
Storage / Metadata
Client
Coordinator
Worker
Connector Plugin
3. Coordinator builds a query plan
Connector plugin provides metadata (table schema, etc.)
Worker

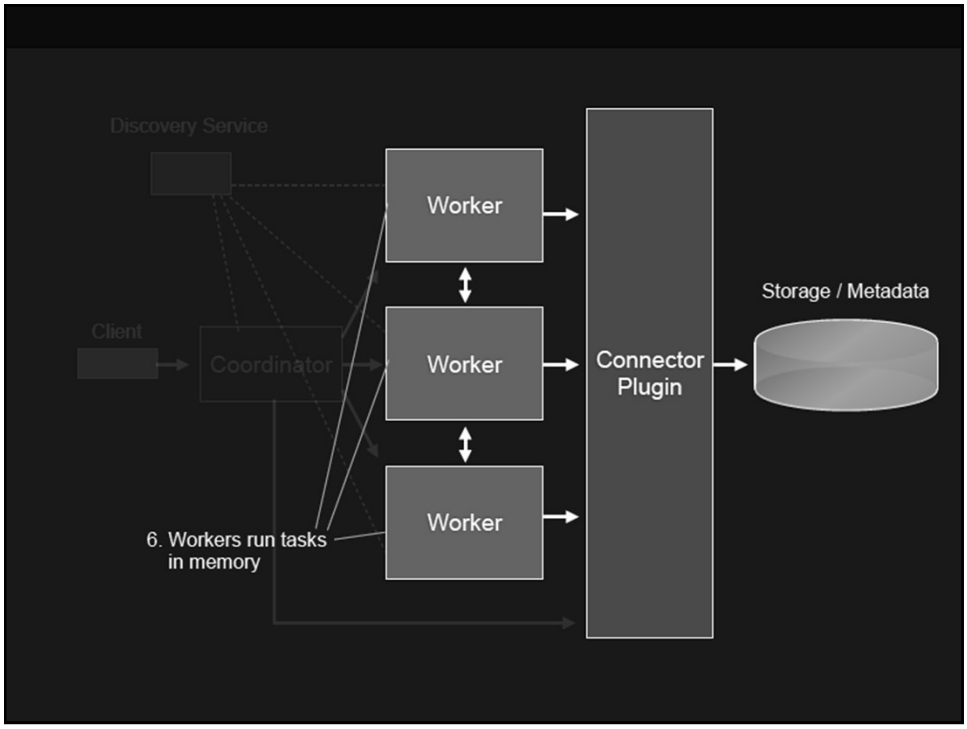
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Discovery Service
Worker
Storage / Metadata
Client
Coordinator
Worker
Connector Plugin
4. Coordinator sends tasks to workers
Worker

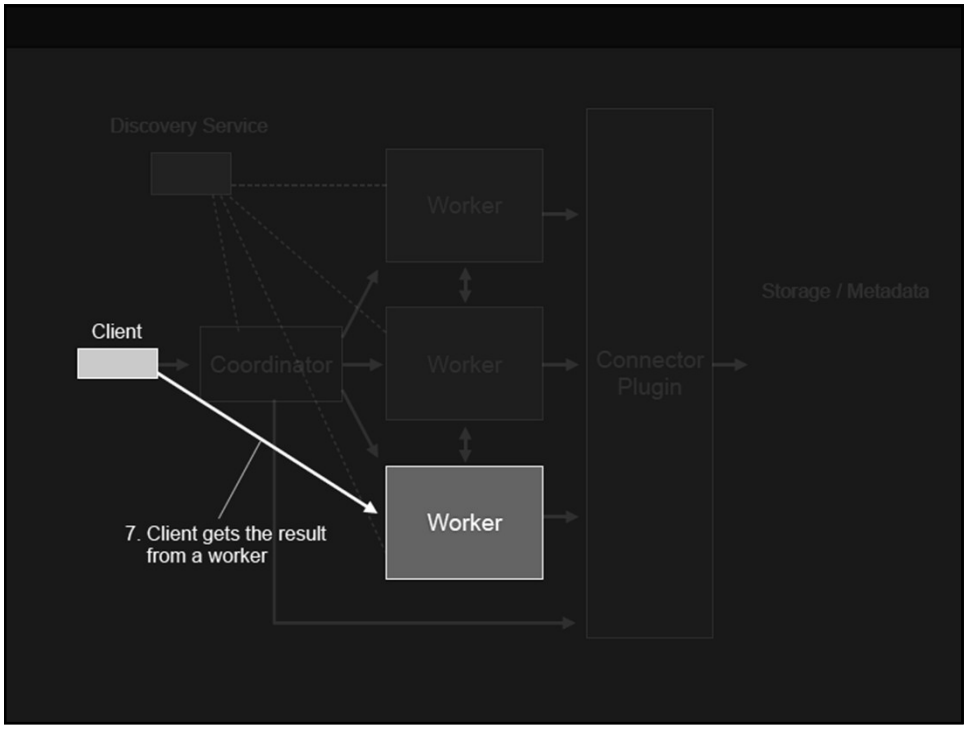
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Discovery Service
Worker
Storage / Metadata
Client
Coordinator
Worker
Connector Plugin
Worker
5. Workers read data through connector plugin

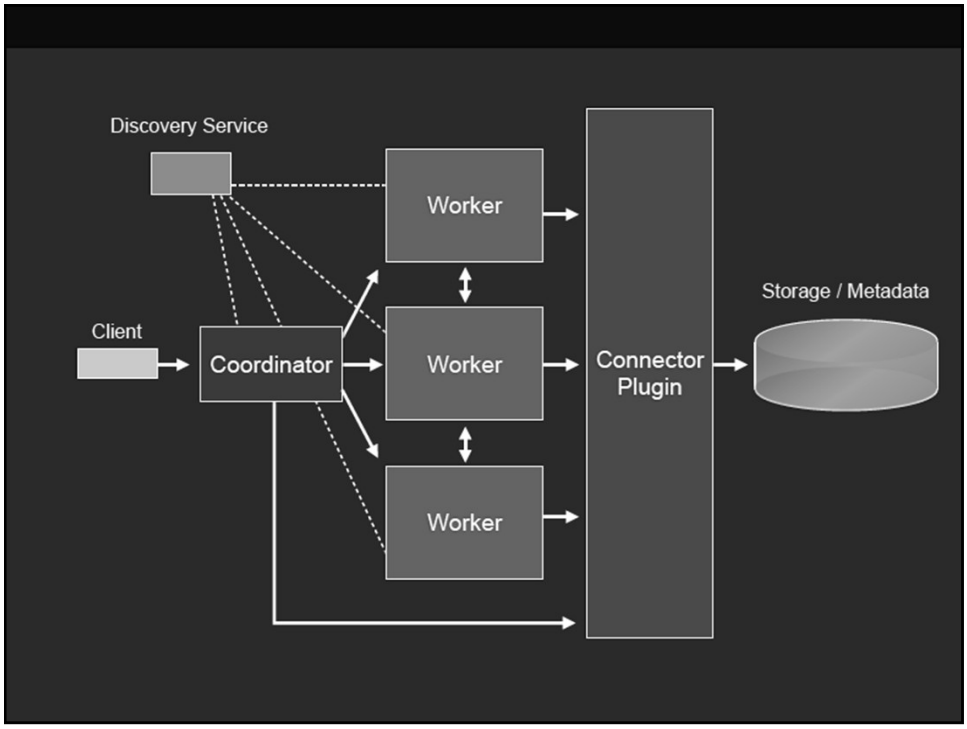
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Discovery Service
Worker
Storage / Metadata
Client
Coordinator
Worker
Connector Plugin
Worker
6. Workers run tasks in memory

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Discovery Service
Worker
Storage / Metadata
Client
Coordinator
Worker
Connector Plugin
Worker
7. Client gets the result from a worker

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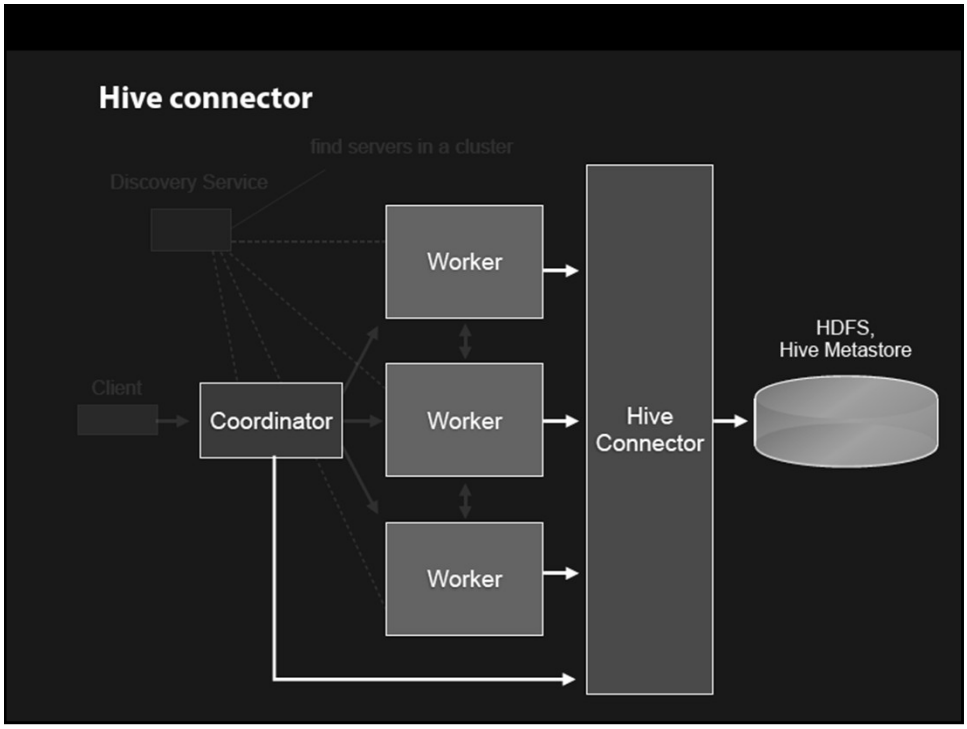

Discovery Service
Worker
Storage / Metadata
Client
Coordinator
Worker
Connector Plugin
Worker

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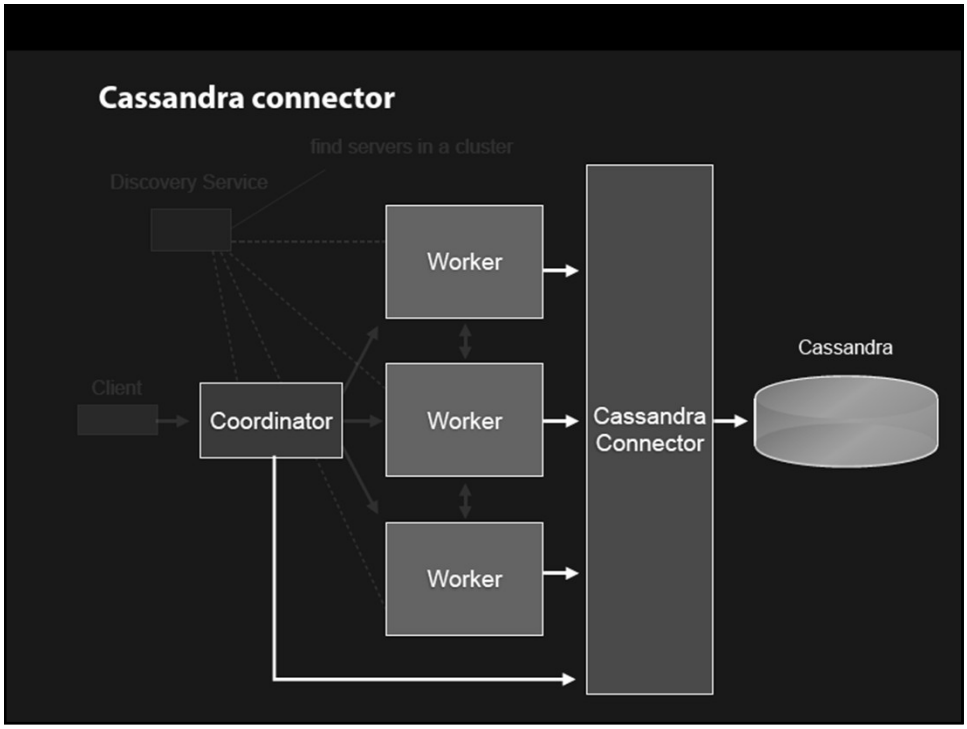
Presto connectors

- Connectors are plugins to Presto
  - written in Java
- Access to storage and metadata
  - provide table schema to coordinators
  - provide table rows to workers
- Implementations
  - Hive connector
  - Cassandra connector
  - MySQL through JDBC connector (prerelease)
  - Or your own connector

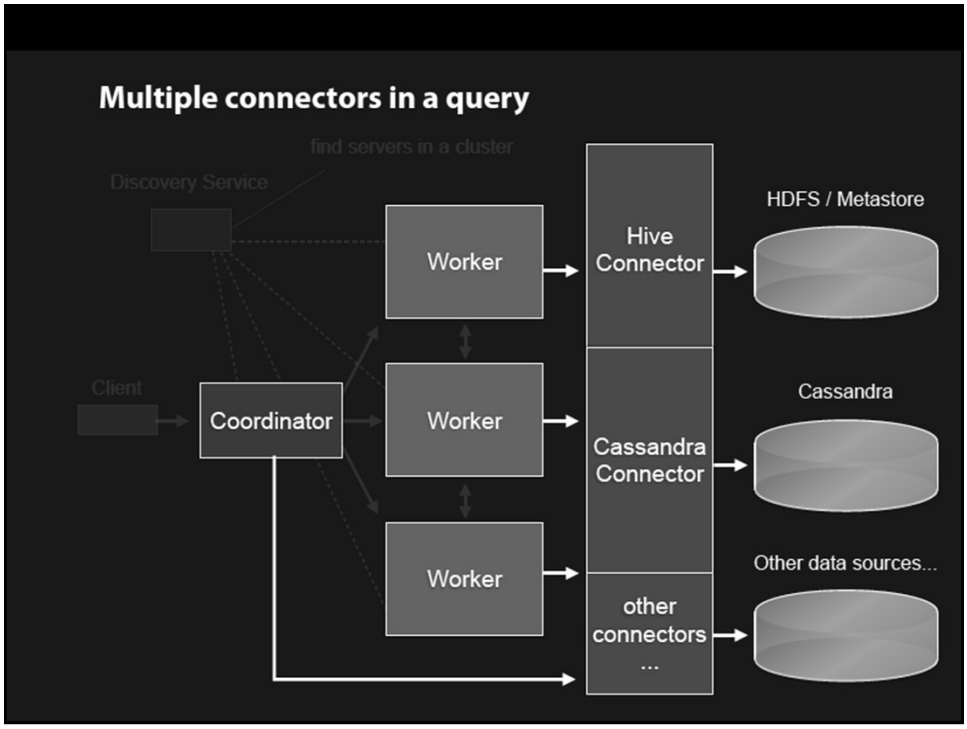
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Cassandra connector
find servers in a cluster
Discovery Service
Client
Coordinator
Worker
Worker
Worker
Cassandra Connector
Cassandra

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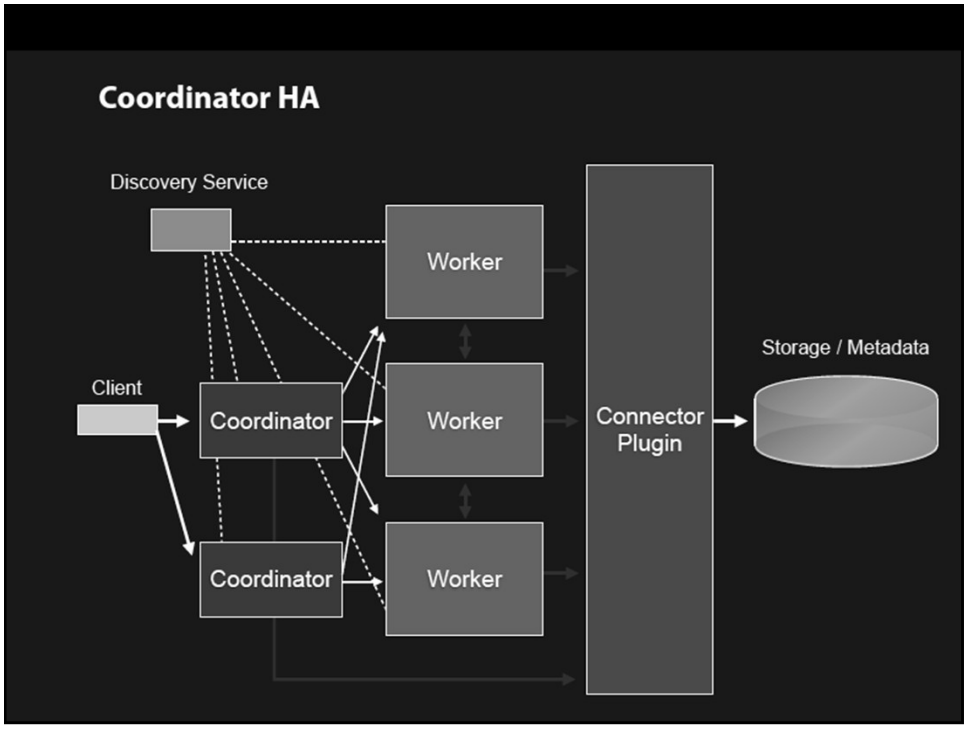

Multiple connectors in a query
find servers in a cluster
Discovery Service
Client
Coordinator
Worker
Worker
Worker
Hive Connector
Cassandra Connector
other connectors ...
HDFS / Metastore
Cassandra
Other data sources...

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Distributed architecture

- 3 type of servers:
  - Coordinator, worker, discovery service
- Get data/metadata through connector plugins.
  - Presto is NOT a database
  - Presto provides SQL to existent data stores
- Client protocol is HTTP + JSON
  - Language bindings: Ruby, Python, PHP, Java (JDBC), R, Node.JS...

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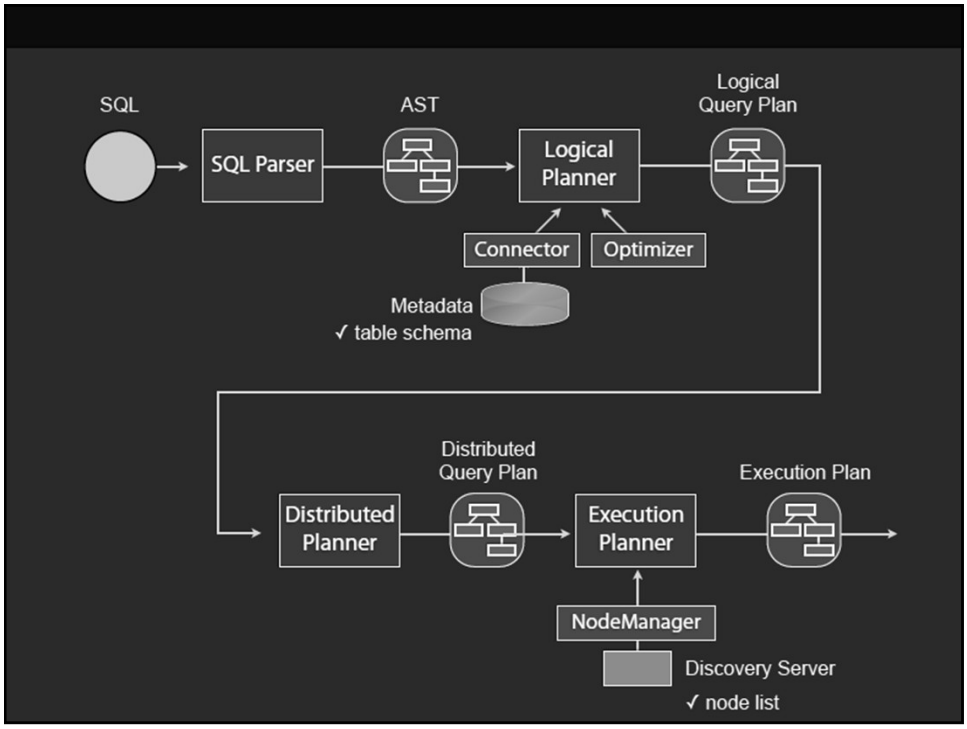


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Presto execution model

- Presto is NOT MapReduce
- Presto's query plan is based on DAG
  - more like Apache Tez or traditional MPP databases
- How query runs?
  - Coordinator
    - SQL Parser
    - Query Planner
    - Execution planner
  - Workers
    - Task execution scheduler

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Query Planner

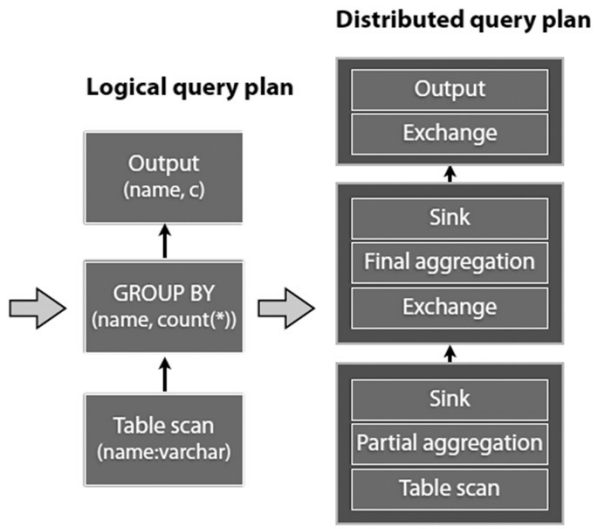
SQL

```
SELECT
  name,
  count(*) AS c
FROM impressions
GROUP BY name
```

+

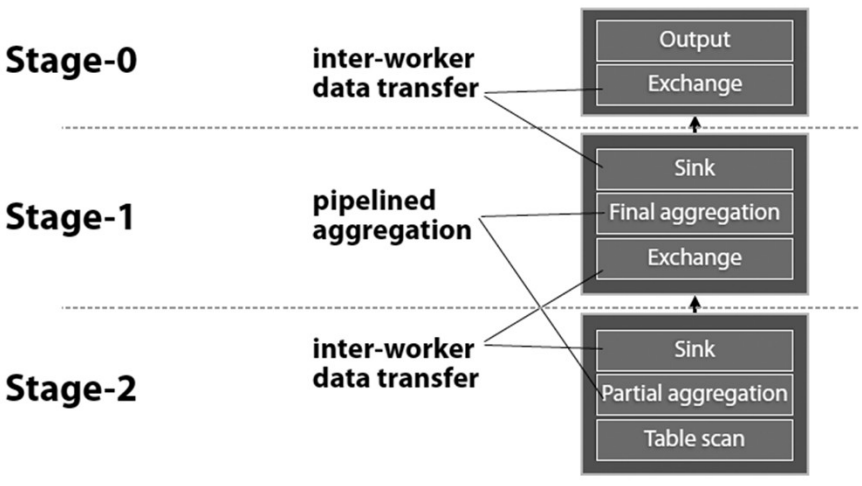
Table schema

```
impressions (
  name varchar
  time bigint
)
```



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Query Planner - Stages

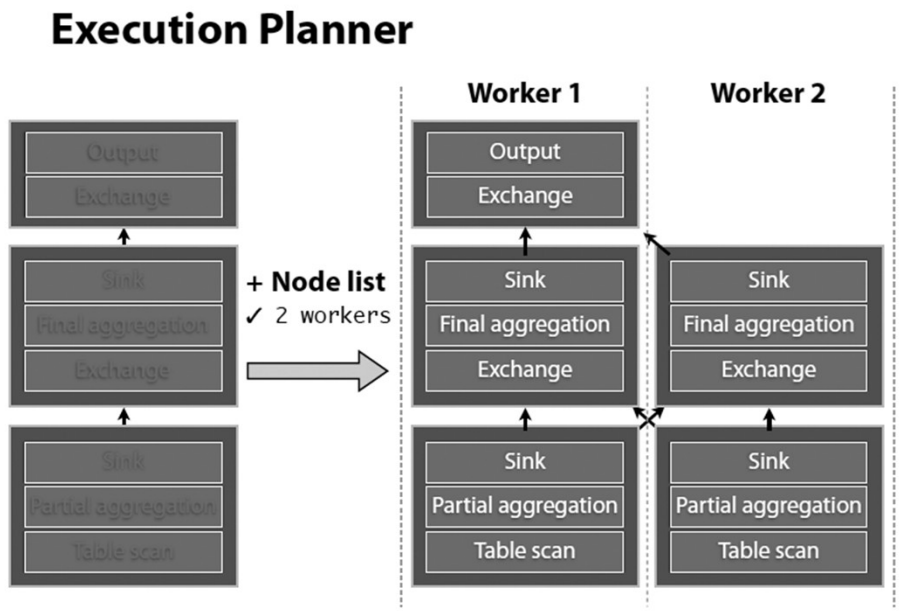


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Stages

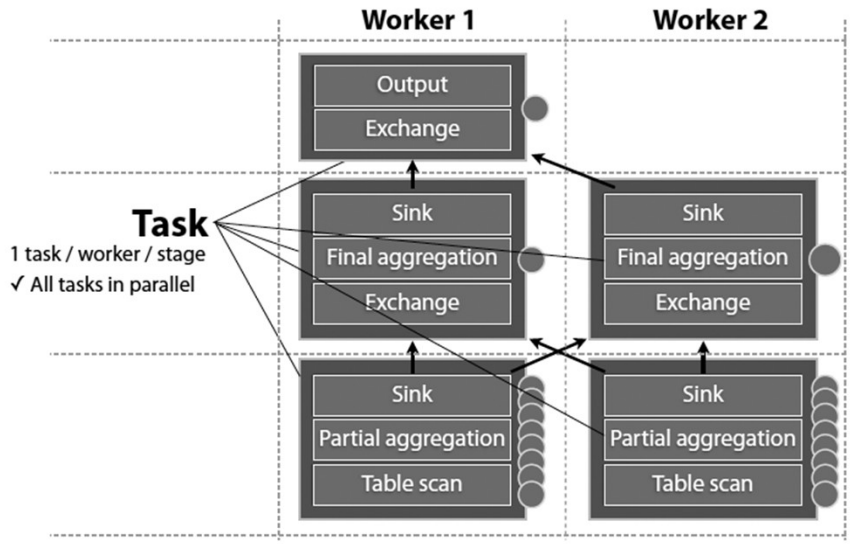
- A stage is a part of the plan that can be executed in parallel across workers
  - Workers execute same computation on different sets of input data
  - Buffered in-memory data transfers (shuffles) between stages to enable data exchange
- Shuffles add latency, use up buffer memory, and have high CPU overhead

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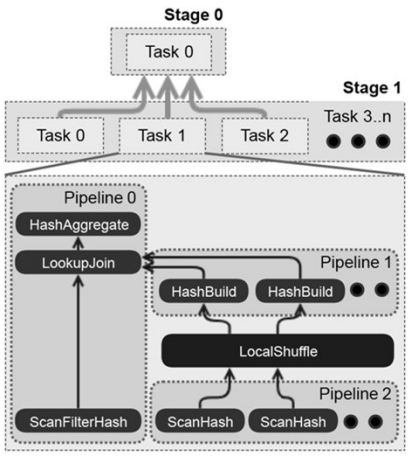
Execution Planner - Tasks



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Tasks

- The coordinator distributes plan stages to workers in the form of executable tasks
- 1 task / 1 worker / 1 stage
- A task may have multiple pipelines
- A pipeline consists of a chain of operators



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Scheduling

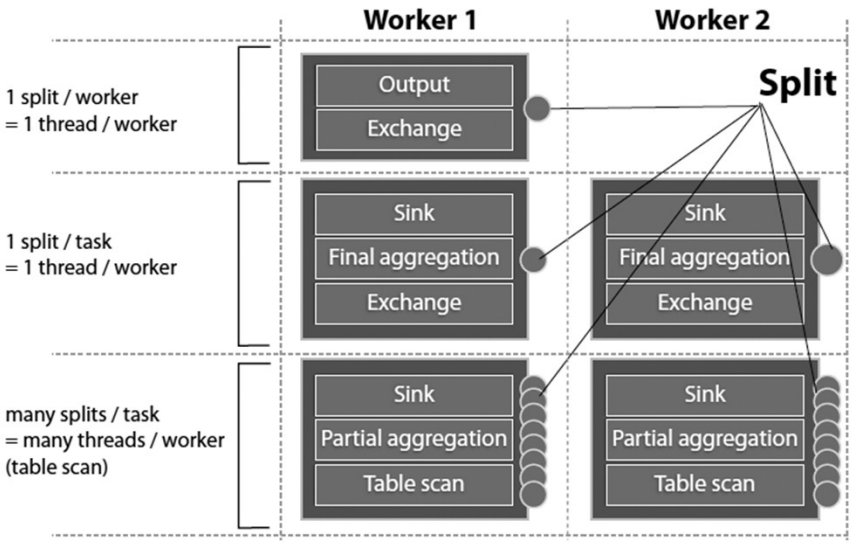
- To execute a query, Scheduler makes two sets of scheduling decisions
- Stage Scheduling
  - all-at-once: minimizes wall clock time by scheduling all stages of execution concurrently
    - data is processed as soon as it is available
    - benefits latency-sensitive use cases such as Interactive Analytics, Developer/Advertiser Analytics, and A/B Testing
  - Phased:
    - Upon a stage should be scheduled according to the policy, it begins to assign tasks for that stage to worker nodes
    - improves memory efficiency for the Batch Analytics use case

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Task scheduling

- Stages are classified into leaf and intermediate stages
- Leaf Stages
  - task scheduler takes into account the constraints imposed by the network and connectors when assigning tasks to worker nodes
    - co-located workers and storage nodes
    - Connector Data Layout constraints
- Intermediate Stages
  - Tasks for intermediate stages can be placed on any worker node
  - The engine still needs to decide how many tasks should be scheduled for each stage

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Split scheduling

- Splits are opaque handles to an addressable chunk of data
  - in an external storage system
  - or intermediate results produced by other workers
  - Eg. Reading from HDFS
    - A split is a file path and offsets to a region of the file
- Split Assignment
  - Splits are assigned to each task lazily
    - Splits are enumerated as the query executes, not up front
    - Queries that can start producing results without processing all the data
    - Splits are assigned to worker with shortest queue
    - Reduces metadata memory usage on coordinator

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Query optimization: Data layouts

- Optimizer takes advantage of physical layout of data
  - Properties: partitioning, sorting, grouping, indexes
- Tables can have multiple layouts with different properties
  - Layouts can have a subset of columns or data
- Optimizer chooses best layout for query
- Tune queries by adding new physical layouts

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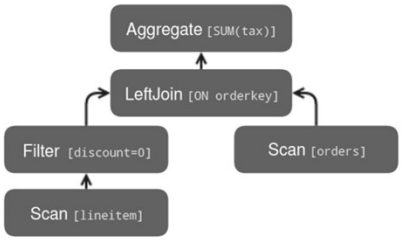
Query optimization: Predicate pushdown

- The optimizer can push range and equality predicates down through the connector improves filtering efficiency
- Engine provides connectors with a two part constraint:
  - Domain of values: ranges and nullability
  - “Black box” predicate for filtering

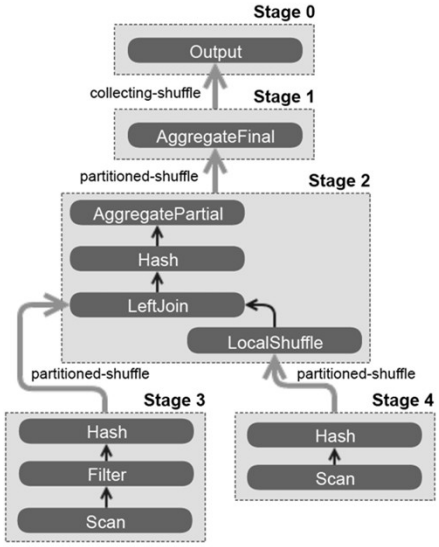
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Eg.

```
SELECT
  orders.orderkey, SUM(tax)
FROM orders
LEFT JOIN lineitem
  ON orders.orderkey = lineitem.orde
WHERE discount = 0
GROUP BY orders.orderkey
```



Logical plan

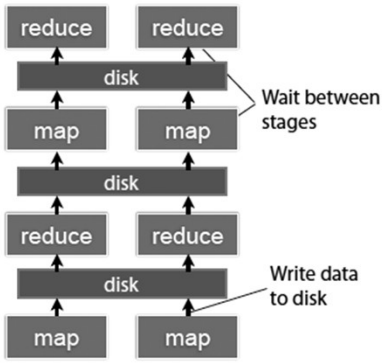


Distributed plan

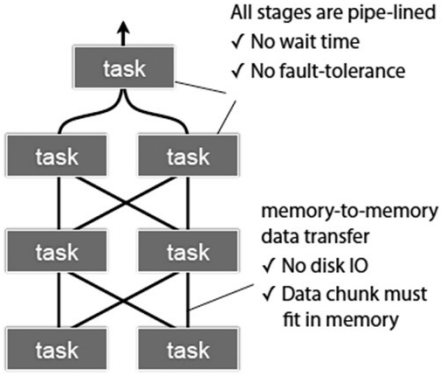
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MapReduce vs. Presto

MapReduce



Presto



All stages are pipe-lined
✓ No wait time
✓ No fault-tolerance

memory-to-memory data transfer
✓ No disk IO
✓ Data chunk must fit in memory

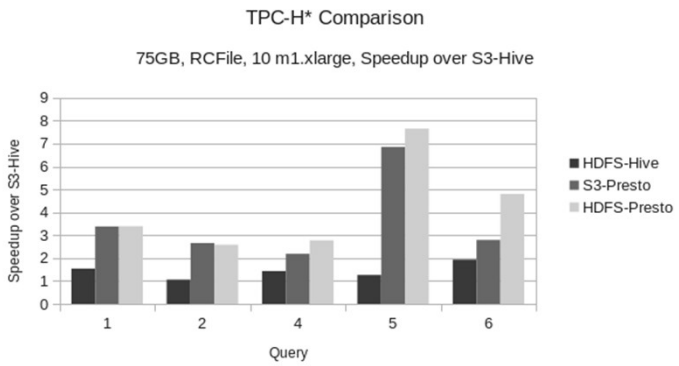
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Query Execution

- SQL is converted into stages, tasks and splits
- All tasks run in parallel
  - No wait time between stages (pipelined)
  - If one task fails, all tasks fail at once (query fails)
- Memory-to-memory data transfer
  - No disk IO
  - If aggregated data doesn't fit in memory, query fails
  - Note: query dies but worker doesn't die. Memory consumption of all queries is fully managed

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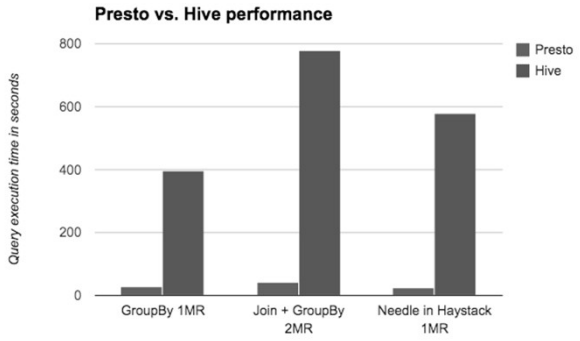
Qubole Presto benchmark



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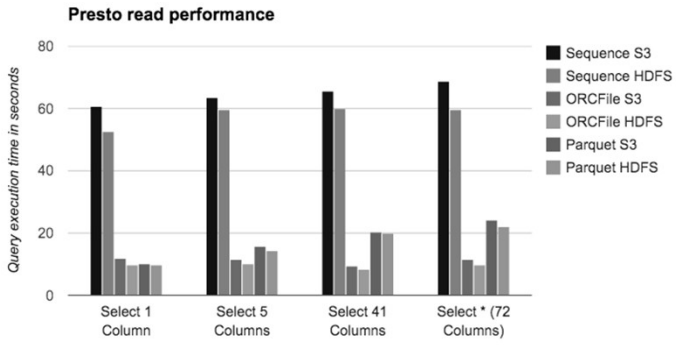
Netflix Presto benchmark (1)

- a group-by query, a join plus a group-by query, and a needle-in-a-haystack (table scan)
- Parquet input files on S3/140GB to 210GB file size
- 40 nodes m2.4xlarge



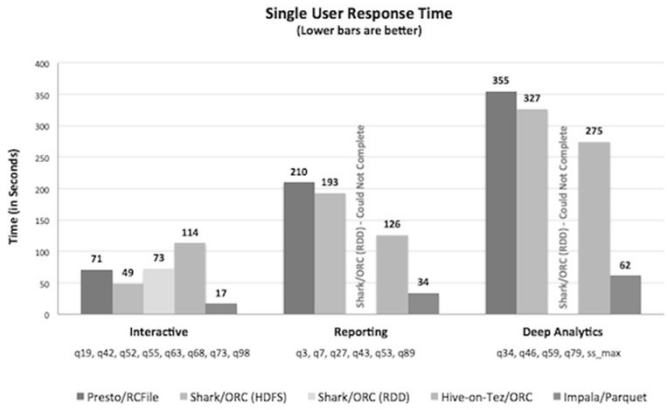
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Netflix Presto benchmark (2)



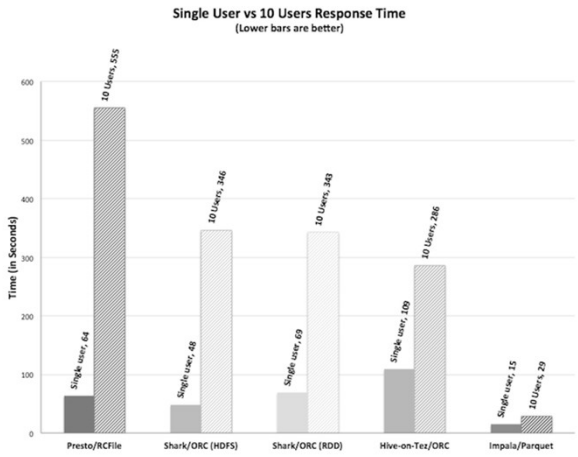
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Cloudera benchmark (1)



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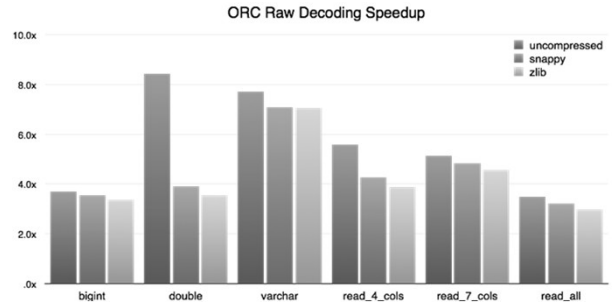
Cloudera benchmark (2)



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Even faster: Data at the speed of Presto ORC

- 6 million rows using the TPC-H
- Read various sets of columns
- Old Hive-based ORC reader vs. the new Presto ORC reader



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References

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